



TRƯỜNG ĐẠI HỌC
BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY
OF SCIENCE AND TECHNOLOGY

NHÓM 1

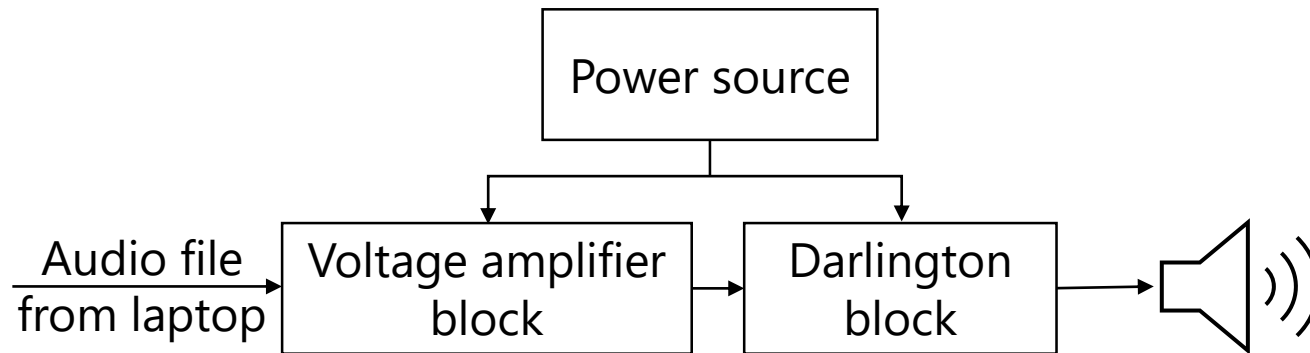
TEAM MEMBERS

- Nguyễn Hồng Hạnh – 20193251
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ONE LOVE. ONE FUTURE.

Build audio amplifier, $V_{cc} = 15V$

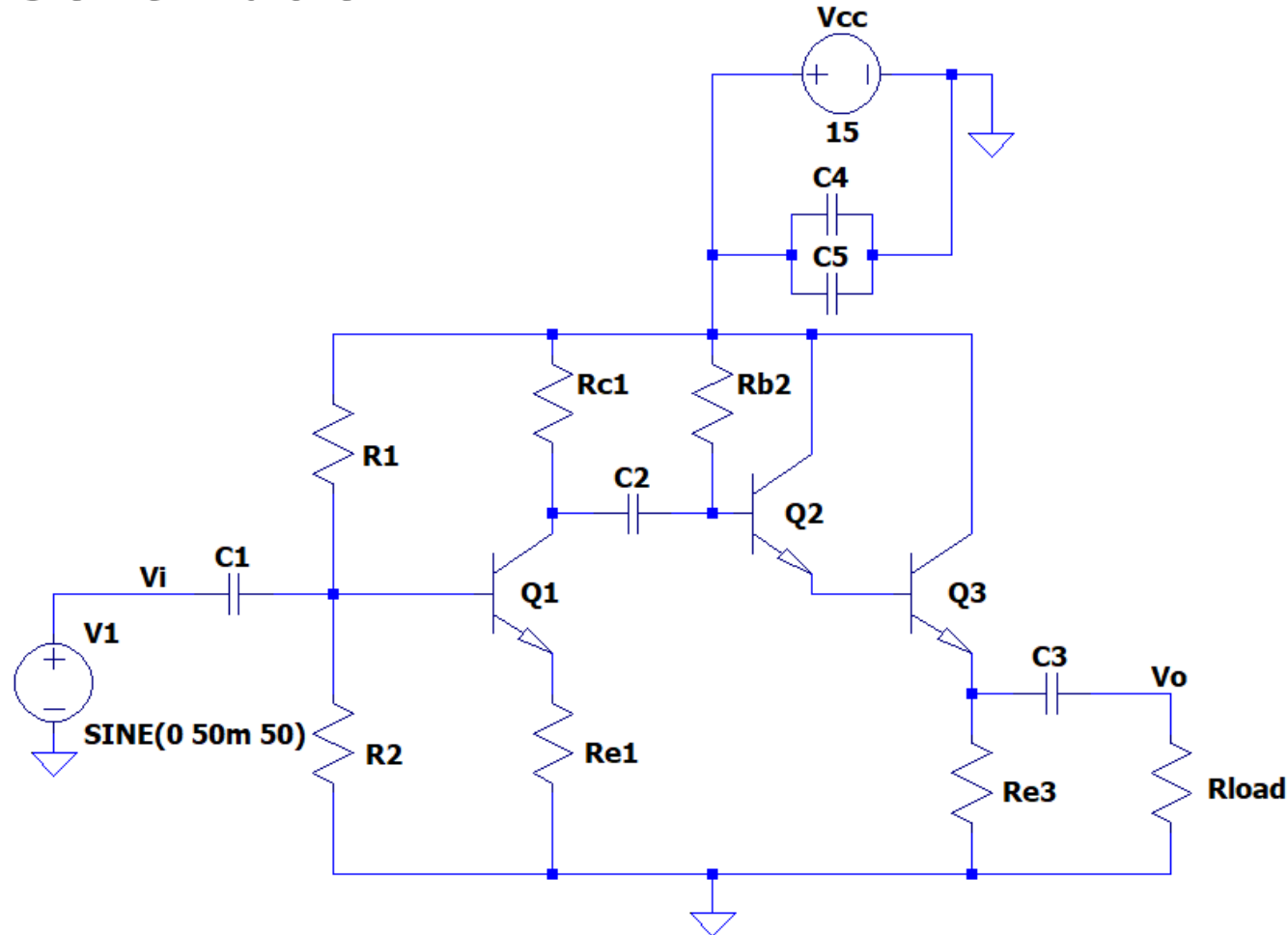
Block diagram



- **Voltage amplifier block:** for amplifying input voltage, the total voltage gain is mainly in this block
- **Darlington block:** to amplify input current and to remove the effect of R load to A_v of the first block

Build audio amplifier, $V_{cc} = 15V$

Schematic



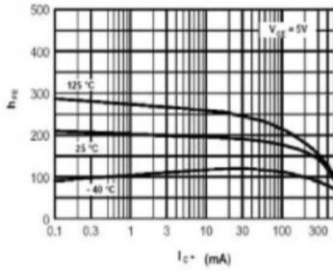
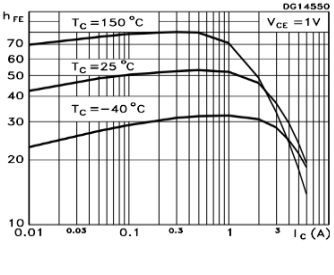
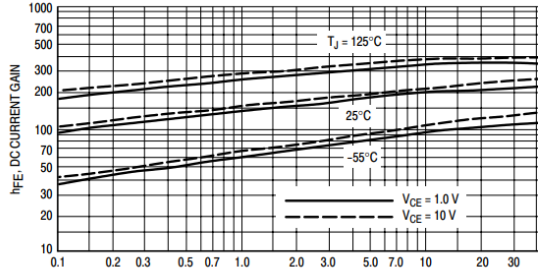
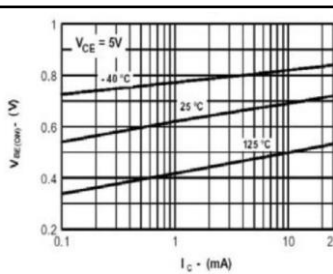
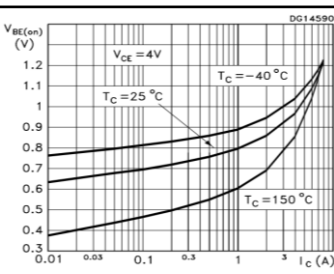
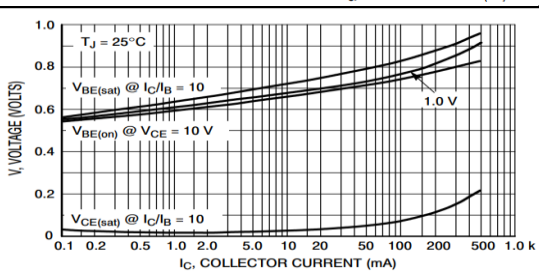
Build audio amplifier, $V_{cc} = 15V$

Specifications

Parameters	Target
Supply voltage	15V
Bandwidth	20Hz – 20kHz
Power out	5W
R_{load}	4 Ω
Voltage gain (A_v)	10

BJT Selection

Comparison:

	Parameters	2N2222	TIP41C	PN2222A
Maximum rating	Vce	30V	100V	40V
	Vcb	60V	100V	75V
	Vbe	5V	5V	6V
	Ic	0.6A	6A	0.6A
Typical characteristic	DC Current Gain			
	Vbe (on)			

Selection

- **Voltage amplifier block (Q1):** 2N2222 BJT because it has stable DC current gain (not change too much when I_C changes) → Easier for calculation and simulation
- **Darlington block:**
 - **Q2:** Collector current in this stage is still small for 2N2222 to bear → 2N2222 for the same reason as above
 - **Q3:** Collector current in this stage is quite large, 2N2222 cannot function well anymore → Employ TIP41C which can function with **BIGGER I_C (6A)**

Darlington block

✓ DC analysis

- From datasheet of 2N2222 and TIP41C, $\beta_2 = 200$, $\beta_3 = 70$

- $I_{c3} = \beta_3 I_{c2} = \beta_2 \beta_3 I_{b2}$

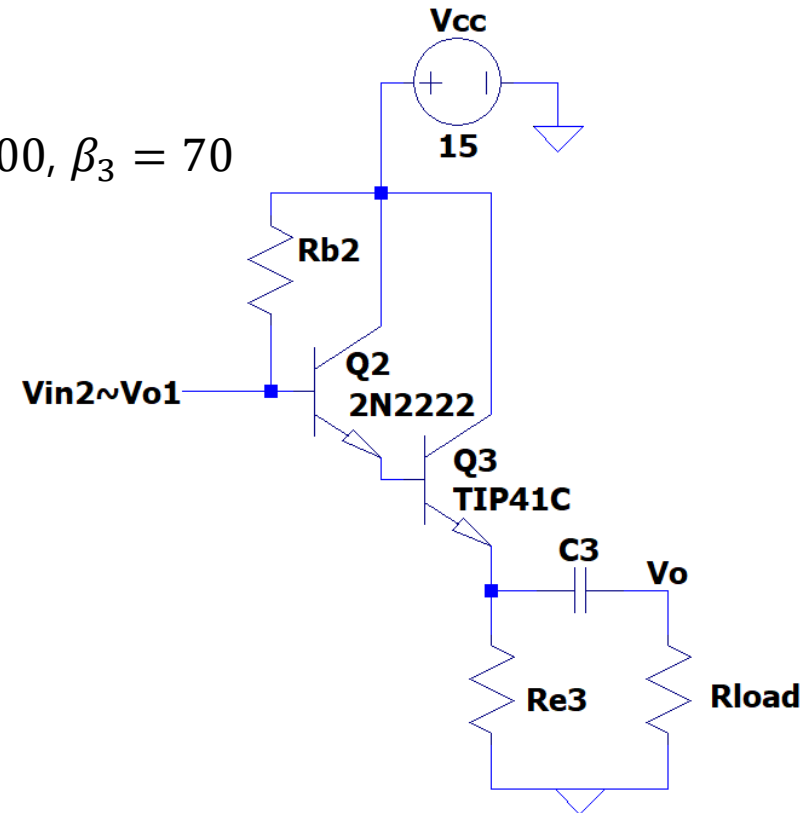
$$15V + I_{b2}R_{b2} + 2(0.65V) + I_{c3}R_{e3} = 0$$

$$\Rightarrow I_{b2} = \frac{13.7V}{R_{b2} + \beta_2 \beta_3 R_{e3}}$$

$$\Rightarrow I_{c3} = \frac{\beta_2 \beta_3 (13.7V)}{R_{b2} + \beta_2 \beta_3 R_{e3}} = \frac{7.5V}{R_{e3}}$$

$$\Rightarrow \text{Choose } R_{e3} = 47\Omega \text{ and } R_{b2} = 510k\Omega$$

$$\Rightarrow \begin{cases} I_{c3} = 0.16A \\ I_{c2} = 2.35mA \end{cases} \Rightarrow \begin{cases} r_{e3} = 0.16\Omega \\ r_{e2} = 11.1\Omega \end{cases}$$

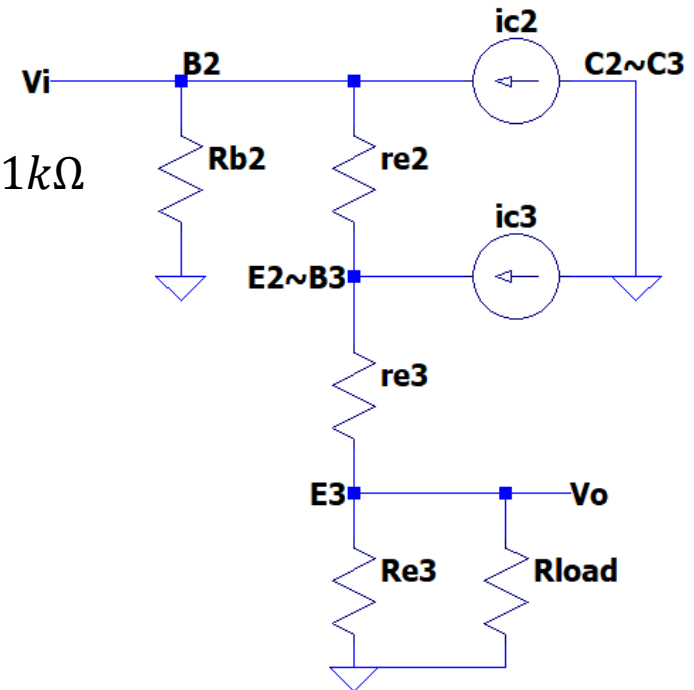


Darlington block

✓ AC analysis

- $$R_{(in\ base\ 2)} = \frac{i_{e2}r_{e2} + i_{e3}(r_{e3} + R_{e3} \parallel R_{load})}{i_{b2}}$$

$$= \beta_2(r_{e2} + \beta_3(r_{e3} + R_{e3} \parallel R_{load})) = 56.1k\Omega$$
- $$A_{i2} = \frac{\beta_2\beta_3R_{b2}}{R_{b2} + R_{(in\ base\ 2)}} = 12682$$
- $$A_{v2} = A_i \frac{R_{e3} \parallel R_{load}}{R_{b2} \parallel R_{(in\ base\ 2)}} = 0.92$$
- $$Z_{in_2} = R_{b2} \parallel R_{(in\ base\ 2)} = 50.5k\Omega$$
- $$Z_{out_2} = R_{e3} \parallel R_{load} \parallel \left(\frac{r_{e2}}{\beta_3} + r_{e3} \right) = 297m\Omega$$



Voltage divider block

✓ DC analysis

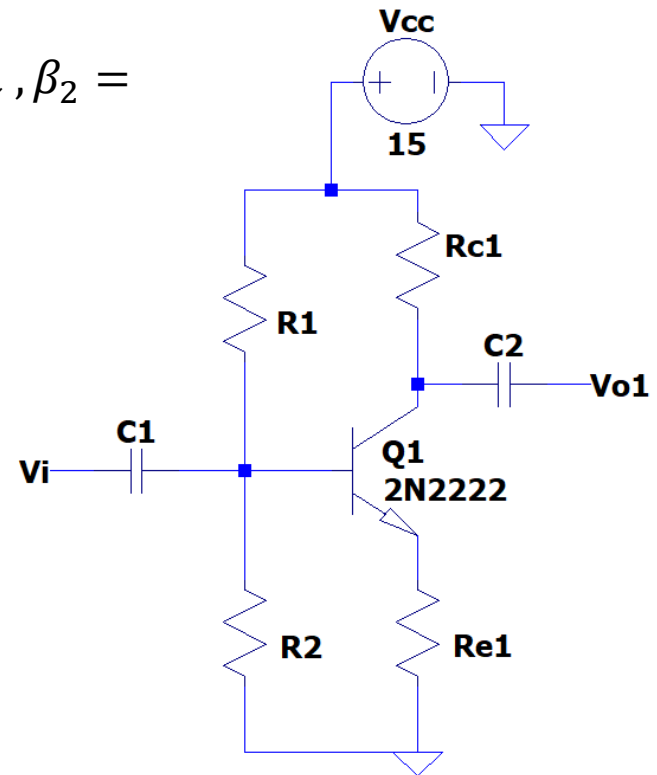
- From datasheet of 2N2222, choose $I_{c1} = 0.185m$, $\beta_2 = 200$, $V_{be} = 0.65V$

$$I_{c1} = \frac{1}{2} I_{c1max} = \frac{7.5V}{R_{c1} + R_{e1}} = 0.185 mA$$

$$\Rightarrow \boxed{R_{c1} + R_{e1} = 40.5k\Omega} \quad (1)$$

- $r_{e1} = \frac{26mV}{I_{c1}} = 141\Omega$

- $$\boxed{I_{c1} = \frac{V_{cc} \cdot \frac{R_2}{R_1 + R_2} - 0.65V}{\frac{R_1 R_L}{\beta_1 (R_1 + R_2)} + R_{e1}}} = 0.185mA \quad (2)$$



Voltage divider block

✓ AC analysis

- $A_{v1}^L = \frac{A_v}{A_{v2}} = \frac{10}{0.92} = 10.87$

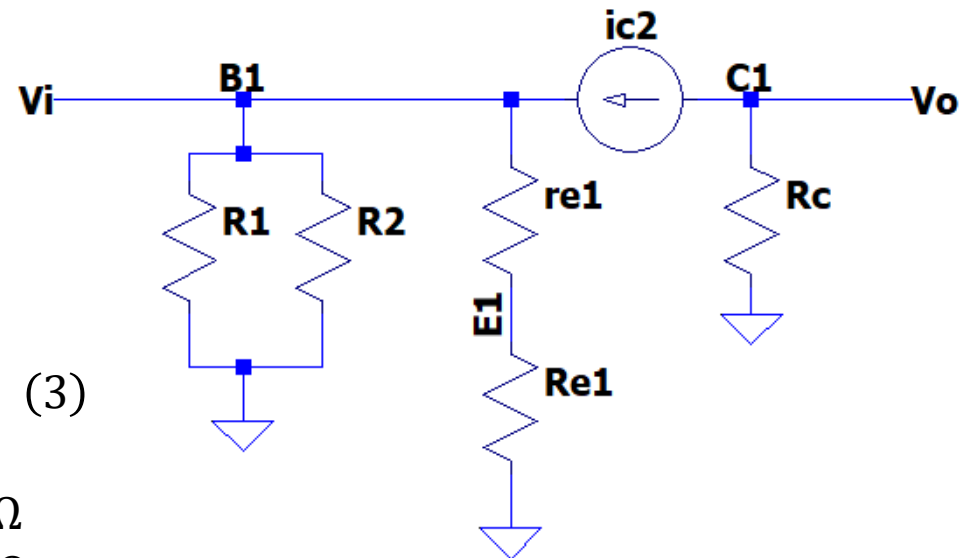
$$\Rightarrow A_{v1}^{NL} = A_{v1}^L \frac{Z_{in2} + R_{c1}}{Z_{in2}}$$

$$\Rightarrow \boxed{\frac{R_{c1}}{141\Omega + R_{e1}} = 10.87 \times \frac{50.5k\Omega + R_{c1}}{50.5k\Omega}} \quad (3)$$

- From (1) and (3), choose $\begin{cases} R_{c1} = 39k\Omega \\ R_{e1} = 1.5k\Omega \end{cases}$

$$\Rightarrow I_c = 0.185mA = \frac{15 \cdot \frac{R_2}{R_1 + R_2} - 0.65V}{\frac{R_1 R_2}{\beta_1 (R_1 + R_2)} + 1.5k\Omega}$$

Choose $R_1 = 51k\Omega \rightarrow R_2 = 3.3k\Omega$



Voltage divider block

✓ AC analysis

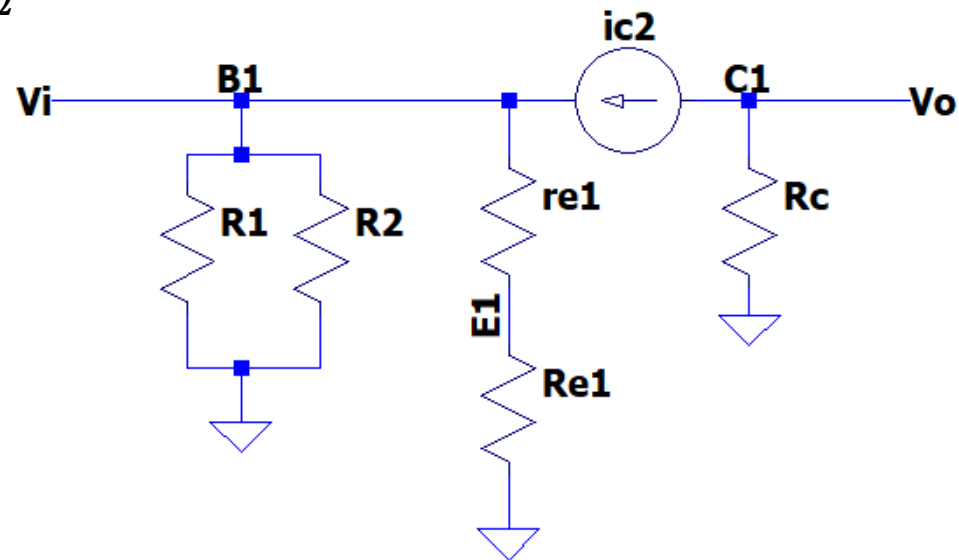
$$\bullet Z_{in_1} = R_1 \parallel R_2 \parallel \beta_1(r_{e1} + R_{e1}) = 3.07k\Omega$$

$$\bullet Z_{out_1} = R_c = 39k\Omega$$

$$\bullet A_{v1}^{NL} = \frac{R_{c1}}{141\Omega + R_{e1}} = 23.8$$

$$\Rightarrow A_{v1}^L = 23.8 \times \frac{Z_{in_2}}{Z_{in_2} + R_{c1}} = 13.4$$

$$\bullet A_{i1}^L = A_{v1}^L \frac{Z_{in_1}}{Z_{in_2}} = 0.81$$



Low frequency respond (high pass filter)

$$\checkmark Z_{in_1} = 3.07k\Omega$$

$$Z_{out_1} = 39k\Omega$$

$$Z_{in_2} = 50.8k\Omega$$

$$Z_{out_2} = 297m\Omega$$

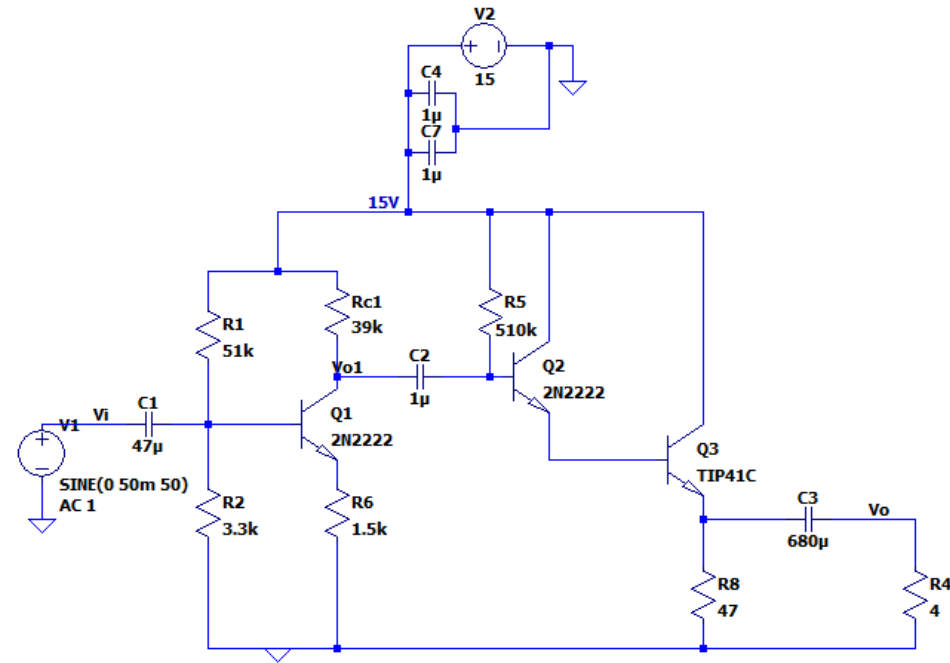
$$\checkmark f_L \leq 20Hz$$

$$\Rightarrow C_1 \geq \frac{1}{2\pi \cdot 20Hz \cdot Z_{in_1}} = 2.6\mu F$$

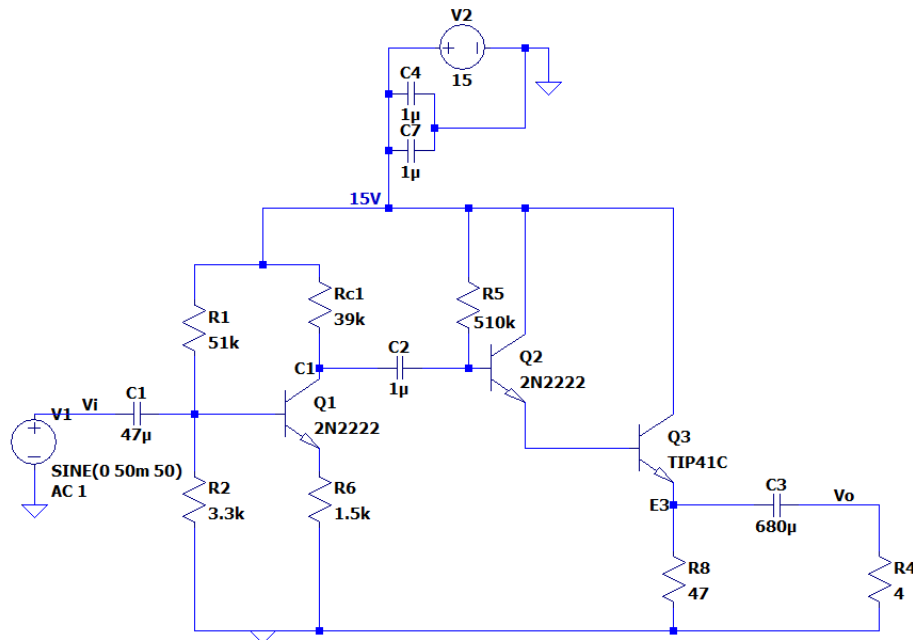
$$C_2 \geq \frac{1}{2\pi \cdot 20Hz \cdot (Z_{out_1} + Z_{in_2})} = 88.6nF$$

$$C_3 \geq \frac{1}{2\pi \cdot 20Hz \cdot (Z_{out_2} + R_{load})} = 1850\mu F$$

$$\Rightarrow \text{Choose } C_1 = 47\mu F, C_2 = 1nF, C_3 = 680\mu F$$



DC simulation



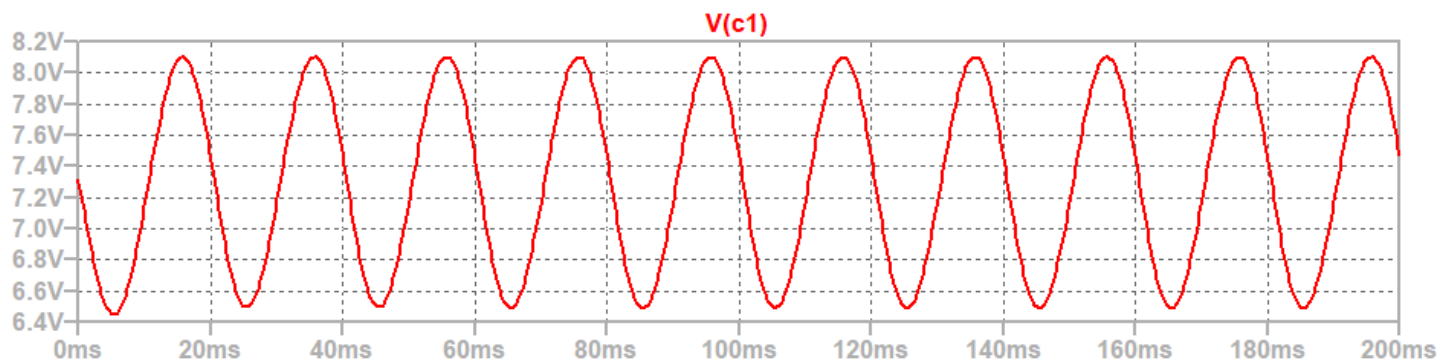
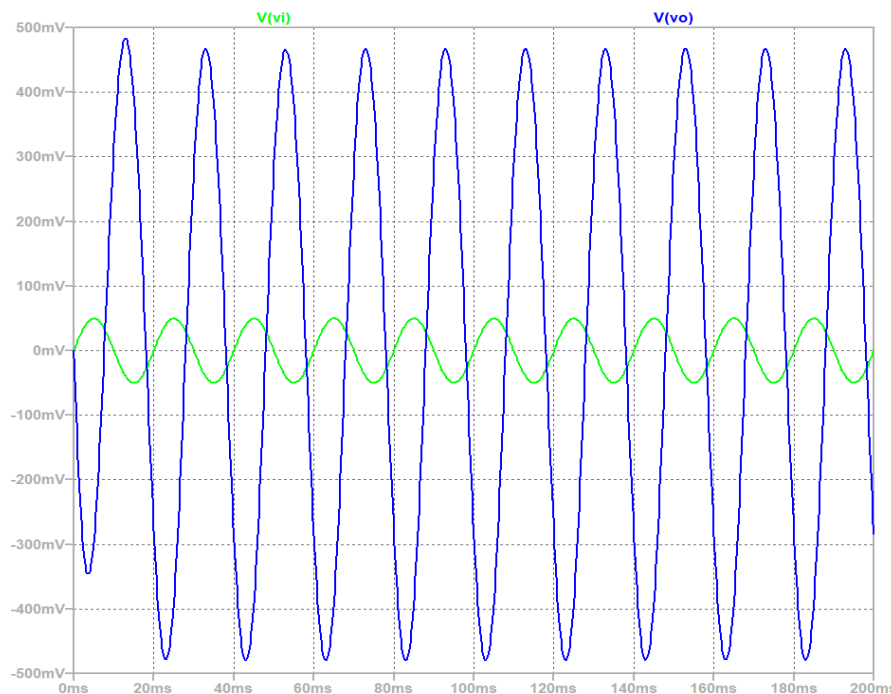
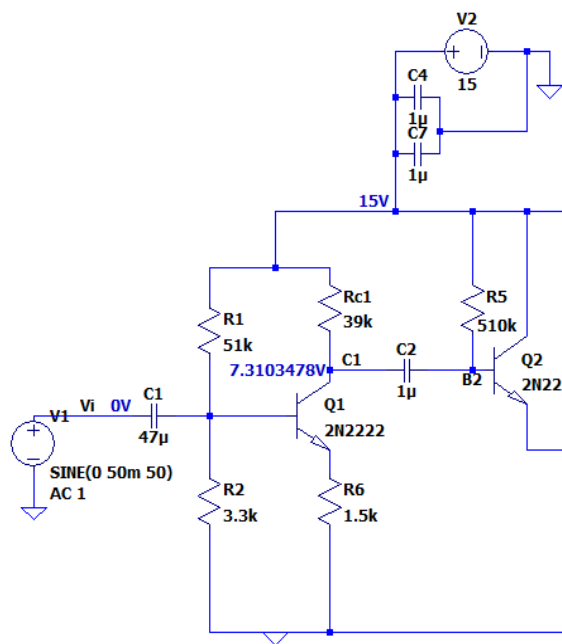
Name :	q3	q2	q1
Model :	tip41c	2n2222	2n2222
Ib :	2.44e-03	1.15e-05	9.27e-07
Ic :	1.63e-01	2.43e-03	1.97e-04
Vbe :	6.37e-01	6.77e-01	6.12e-01
Vbc :	-6.57e+00	-5.89e+00	-6.40e+00
Vce :	7.20e+00	6.57e+00	7.01e+00
BetaDC :	6.70e+01	2.10e+02	2.13e+02

--- Operating Point ---

V(c1) :	7.31035	voltage
V(n003) :	0.908729	voltage
V(n005) :	0.297147	voltage
V(n001) :	15	voltage
V(vi) :	0	voltage
V(n002) :	9.11164	voltage
V(n004) :	8.43427	voltage
V(e3) :	7.79713	voltage
V(vo) :	2.12082e-014	voltage
Ic (Q3) :	0.163458	device_current
Ib (Q3) :	0.00243811	device_current
Ie (Q3) :	-0.165896	device_current
Ic (Q2) :	0.00242658	device_current
Ib (Q2) :	1.15458e-005	device_current
Ie (Q2) :	-0.00243813	device_current
Ic (Q1) :	0.000197171	device_current
Ib (Q1) :	9.27106e-007	device_current
Ie (Q1) :	-0.000198098	device_current
I (C7) :	-1.5e-017	device_current
I (C4) :	-1.5e-017	device_current
I (C3) :	-5.30205e-015	device_current
I (C2) :	1.80129e-018	device_current
I (C1) :	4.27103e-017	device_current
I (R8) :	0.165896	device_current
I (R6) :	0.000198098	device_current
I (R4) :	5.30205e-015	device_current
I (R5) :	1.15458e-005	device_current
I (R2) :	0.000275372	device_current
I (R1) :	0.000276299	device_current
I (Rc1) :	0.000197171	device_current
I (V1) :	4.27103e-017	device_current
I (V2) :	-0.16637	device_current

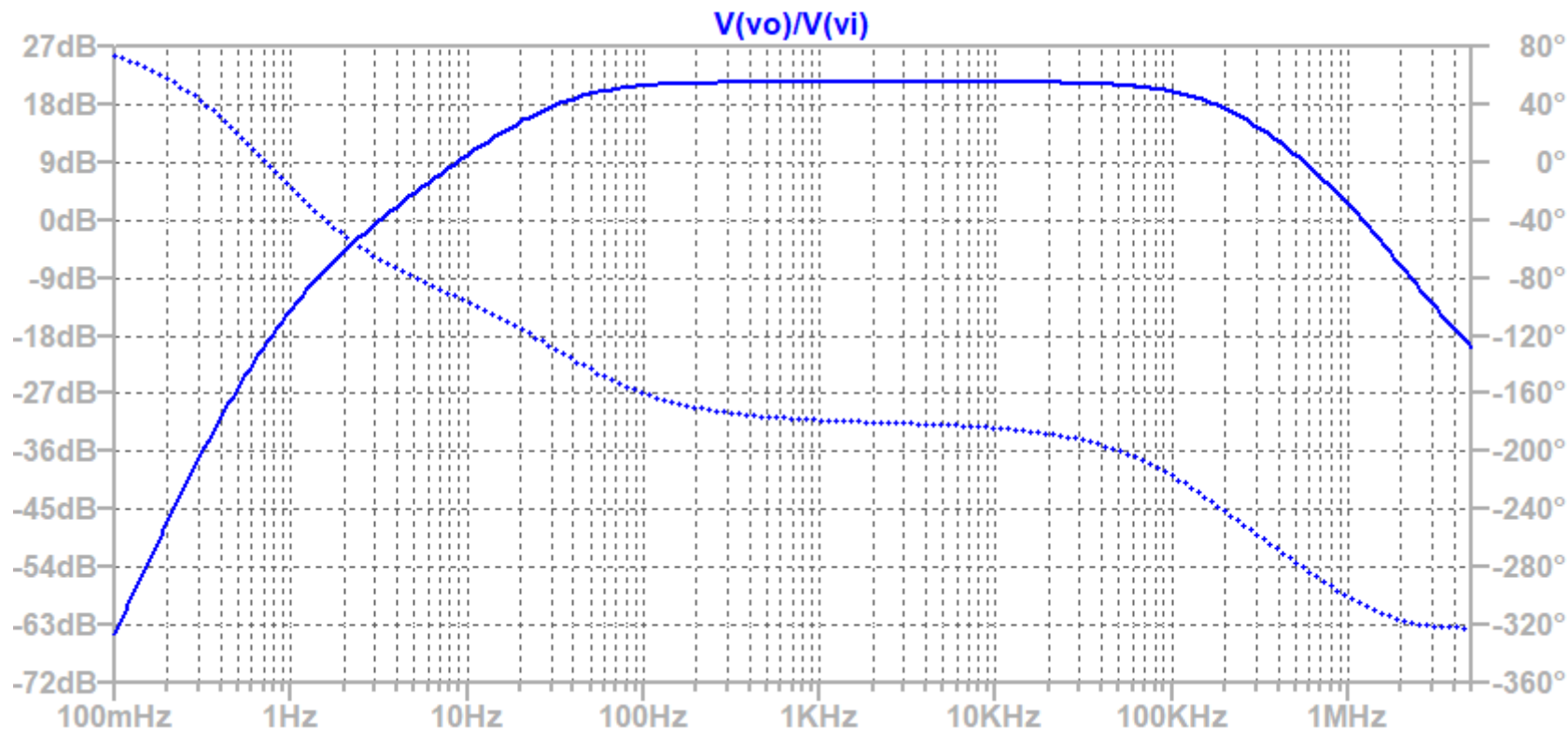
Simulation

AC simulation

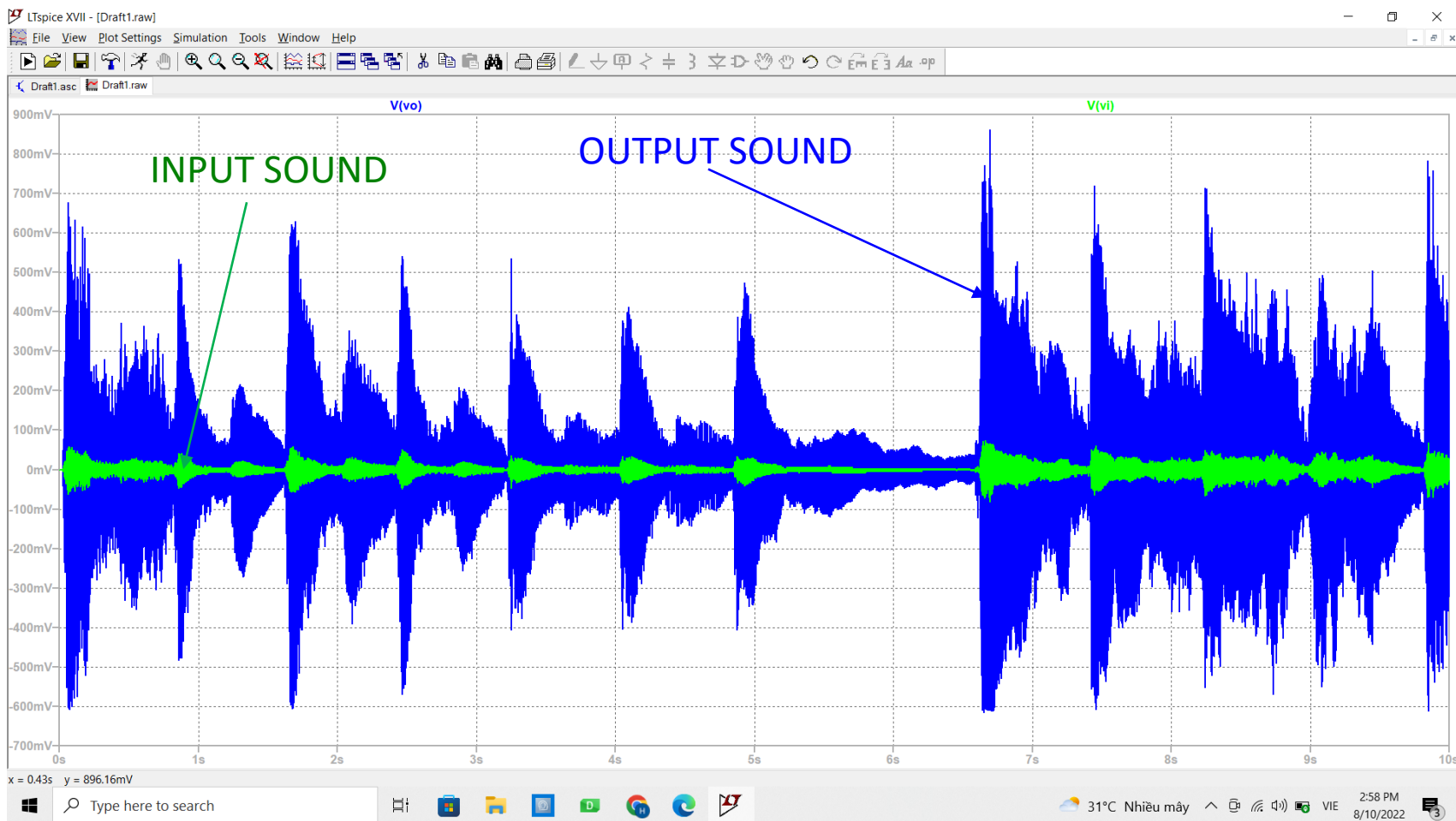


Simulation

Cutoff frequency



Sound simulation

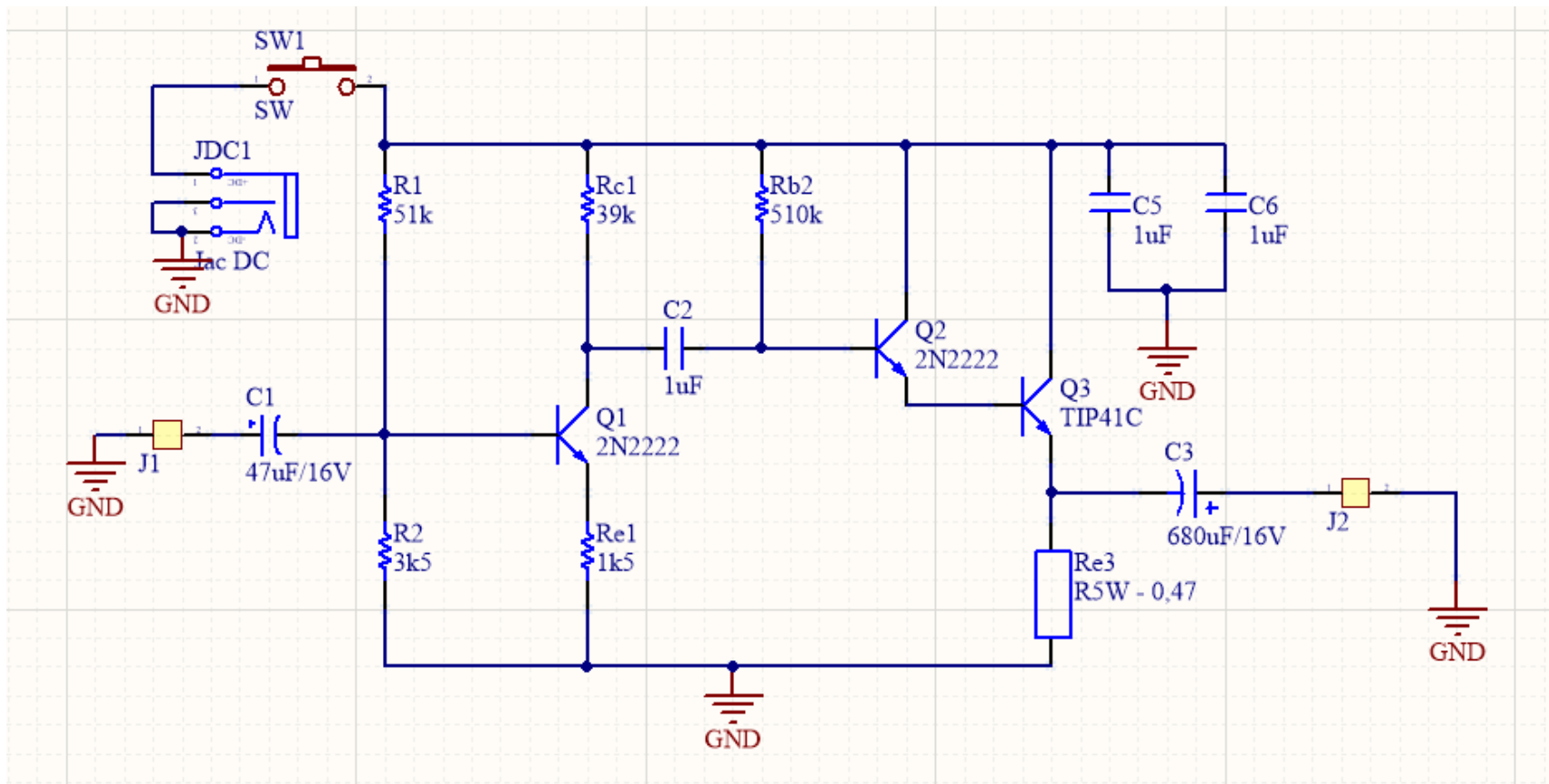


Calculation vs. Simulation Results

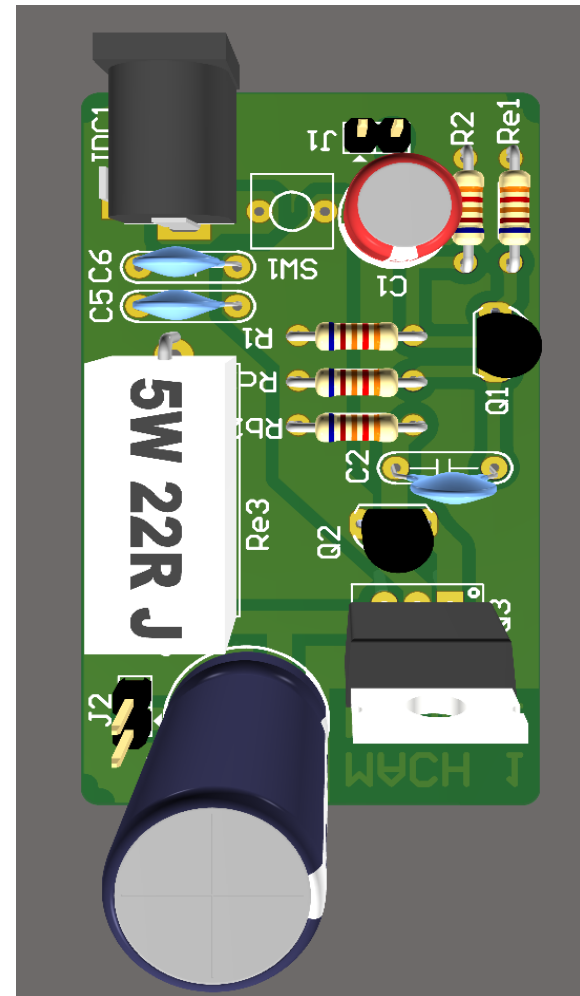
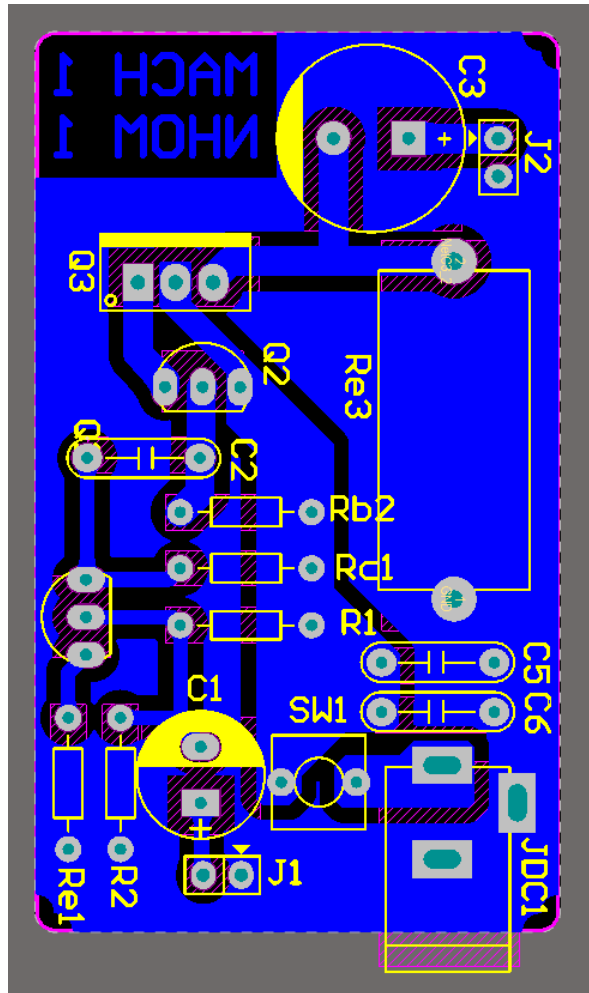
Summary table

Parameters	Calculation	Simulation
β_1	200	213
β_2	200	210
β_3	70	67
I_{c1}	0.185mA	0.197mA
V_{c1}	7.79V	7.31V
I_{c3}	160mA	163mA
V_{e3}	7.52V	7.80
A_{v1}^L	13.4	15.8
A_{i1}^L	0.81	0.68
A_{v2}	0.92	0.60
A_{i2}	12682	12300
A_v	12.3	9.48
A_i	10272	8364
Bandwidth	20Hz - 20kHz	35Hz - 159kHz

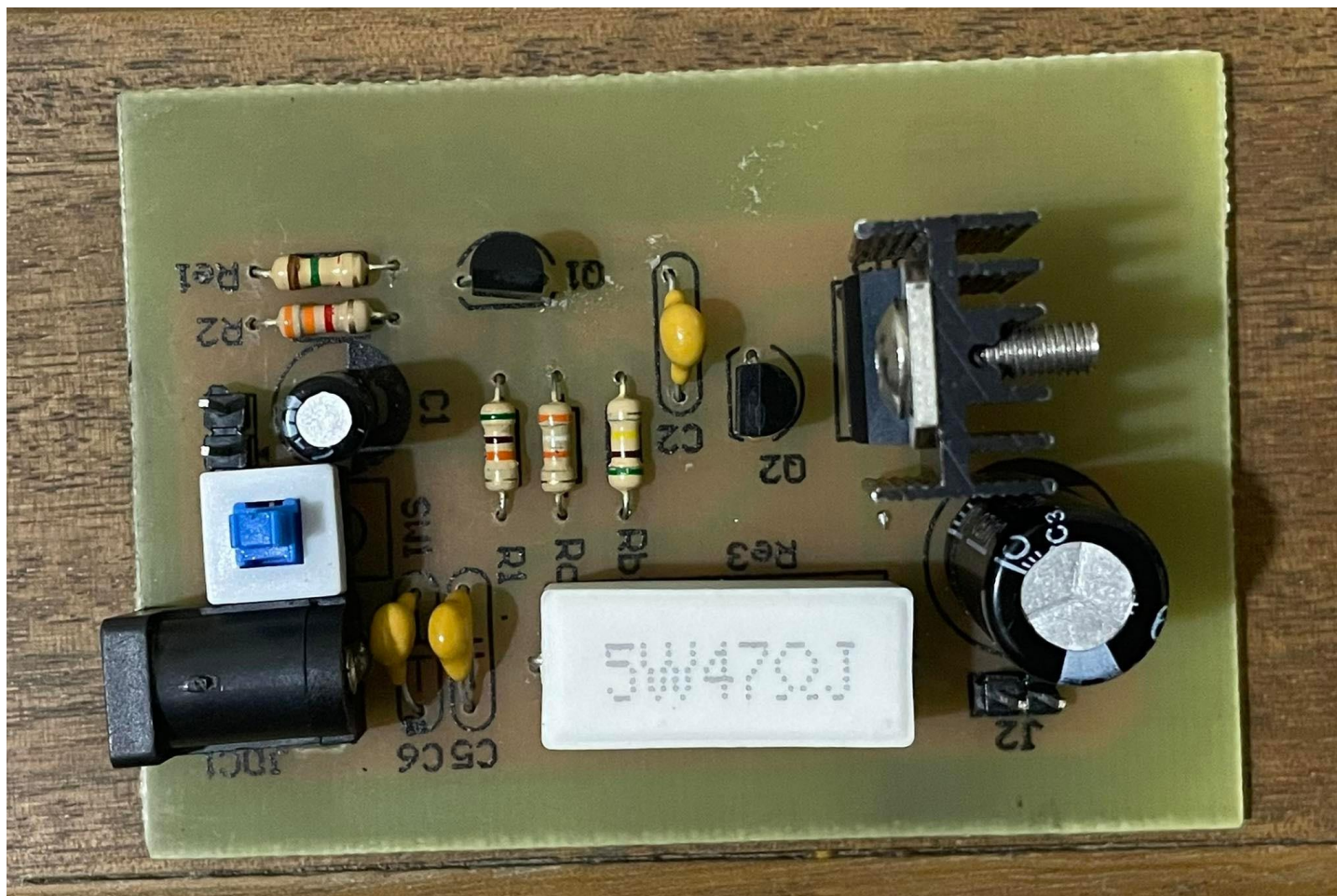
Schematic



PCB

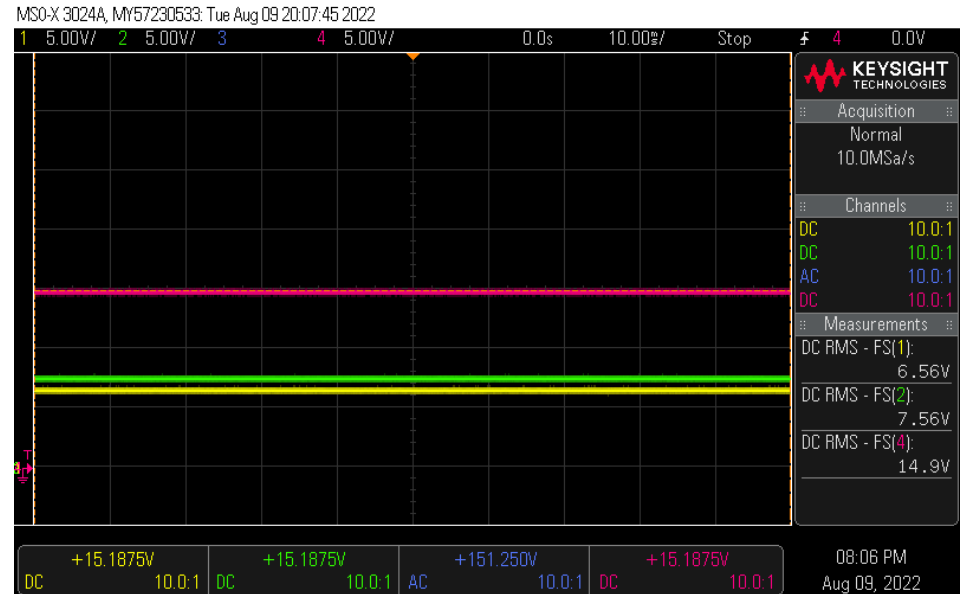
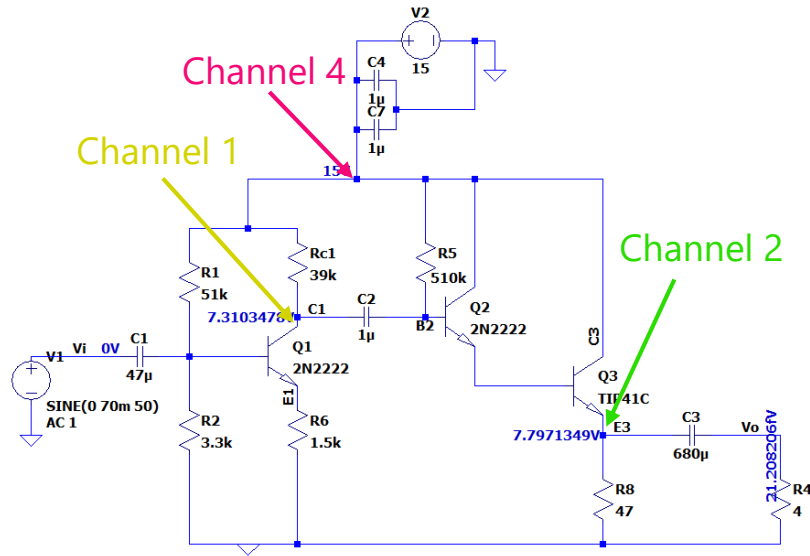


Final product



Final product evaluation

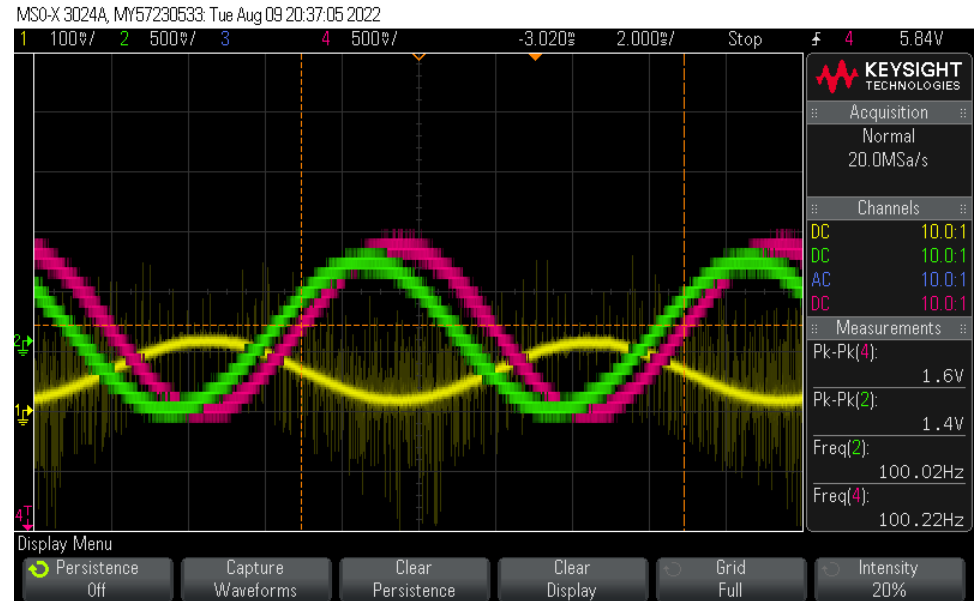
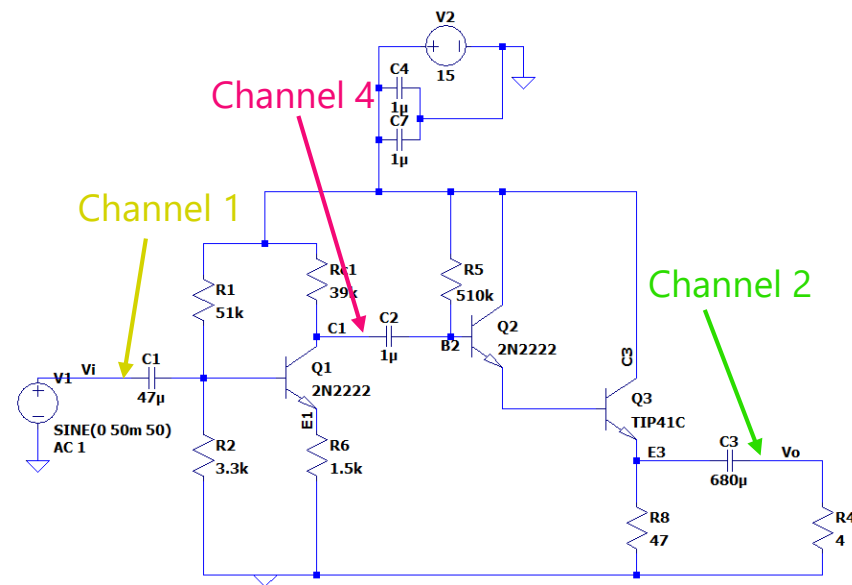
Measure Q point



- $V_{CC} = 14.9V$
- $V_{C1} = 6.56V$
- $V_{E3} = 7.56V$

Final product evaluation

AC Measurement results with Oscilloscope



- $v_{in} = 100mV_{pp}, f = 100Hz$
- $v_{c1} = 1.6V_{pp}, f = 100Hz$
- $v_{e3} = 1.4V_{pp}, f = 100Hz$

$$\Rightarrow \begin{cases} A_{V1}^L = 16 \\ A_{v2} = 0.875 \\ A_v = 14 \end{cases}$$

A large, stylized graphic on the left side of the slide. It consists of a red background with a circular pattern of small, light red dots that form a larger, abstract shape resembling a stylized 'H' or a series of concentric arcs. The word 'HUST' is written in white, bold, sans-serif capital letters across the center of this graphic.

HUST

THANK YOU !